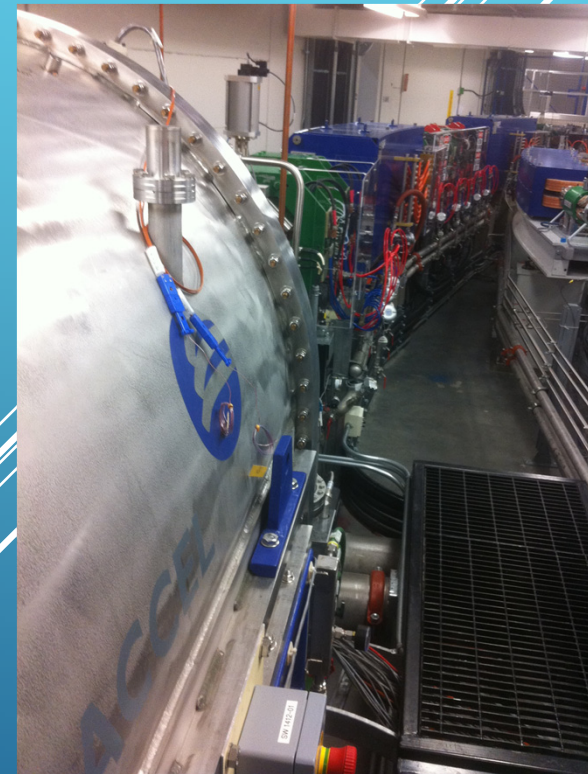


SUPERCONDUCTING RF SYSTEM PLANS AT CLS

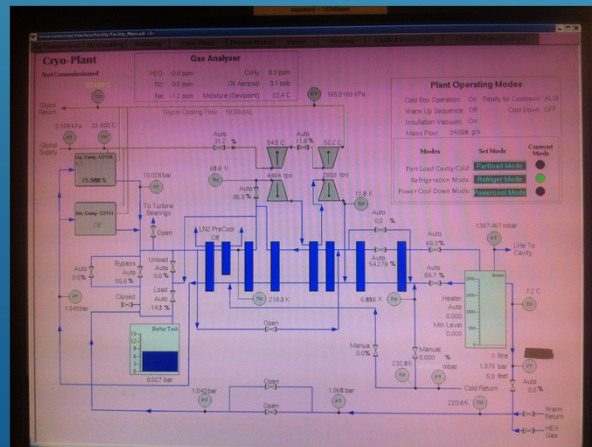
Chris Regier, Ph.D., P.Eng.

Mechanical/Cryogenic Engineer



CRYOGENICS AT CLS - OUTLINE

- ▶ The Canadian Light Source
- ▶ The storage ring SRF system
 - ▶ The CESR-B cryomodule
- ▶ CESR-B history at CLS
- ▶ Past problems
- ▶ The “Cryo Upgrade”
- ▶ Superconducting vs. normal-conducting
- ▶ Another CESR-B for CLS
- ▶ The cold RF test



THE CANADIAN LIGHT SOURCE

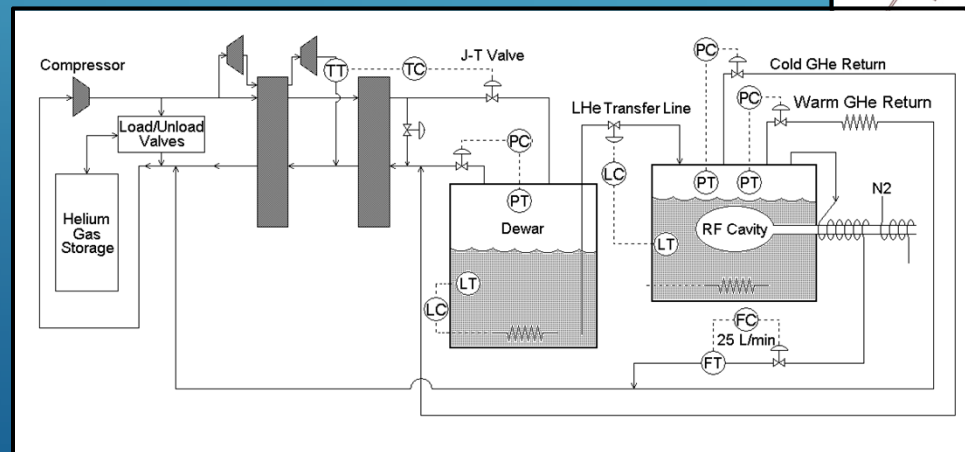
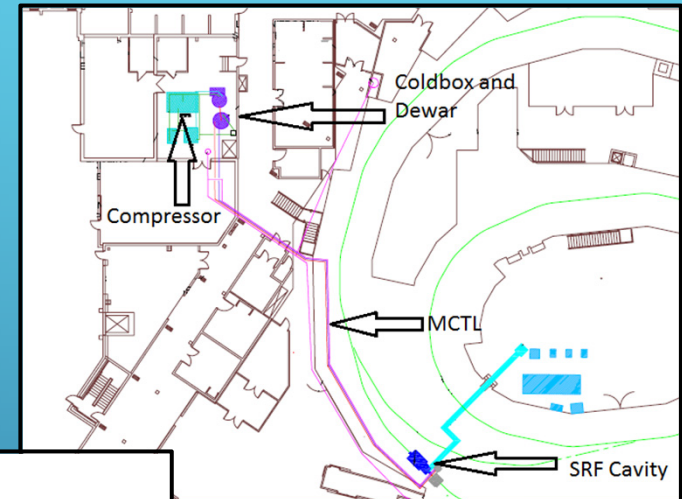
- ▶ CLS is a 3rd Generation Synchrotron
 - ▶ Canada's only synchrotron
 - ▶ Located in Saskatoon, Canada
 - ▶ Population 250,000
 - ▶ Western Canadian prairie
 - ▶ ~1200 km to Vancouver, ~2200 km to Toronto
- ▶ 13 active beamlines + 4 under construction
- ▶ CLS uses superconducting RF to replenish energy lost to synch radiation
 - ▶ Storage ring was commissioned with SRF



THE STORAGE RING SRF SYSTEM

► Major Components

- Linde TCF-50 coldbox
- Kaeser ESD351 compressor
- 2000 L dewar
- ~55 m MCTL to valve box above cryomodule
- 157 m³ buffer
- Single SRF module in storage ring

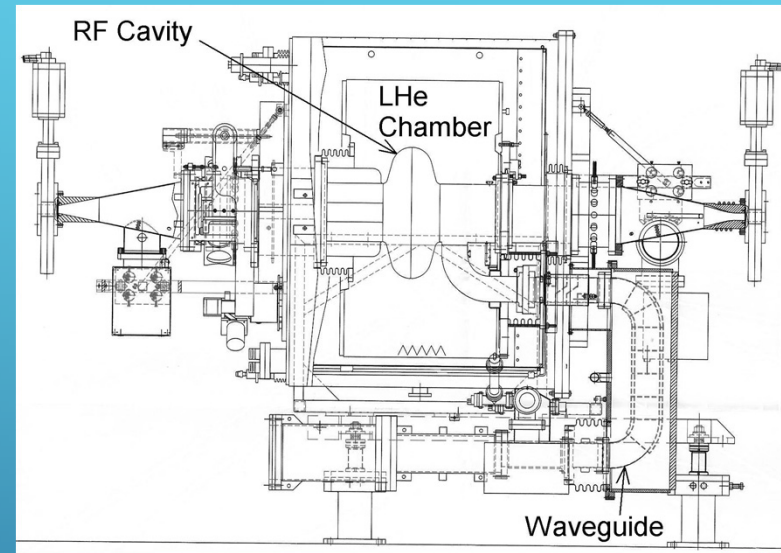


THE STORAGE RING SRF SYSTEM



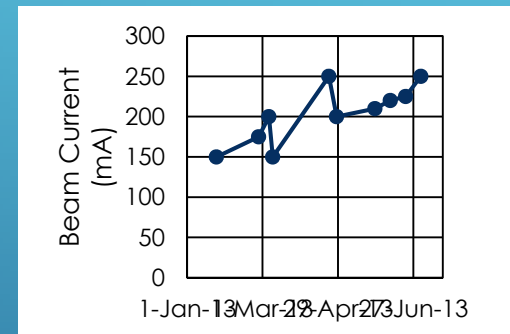
THE STORAGE RING SRF SYSTEM

- ▶ 500 MHz CESR-B elliptical SRF cavity
 - ▶ CLS has 2 units (one operating, one offline spare)
- ▶ Designed at Cornell in 1990s
- ▶ Constructed by Accel (now RI)
- ▶ $V_{\text{acc}} = 2.05 \text{ MV}$
- ▶ 310 kW Thales TB 2161 klystron
- ▶ Other facilities operating the CESR-B
 - ▶ Cornell (CHESS)
 - ▶ TLS
 - ▶ Diamond
 - ▶ SSRF
 - ▶ PLS
 - ▶ NSLS-II
 - ▶ SIRIUS (planning in progress)



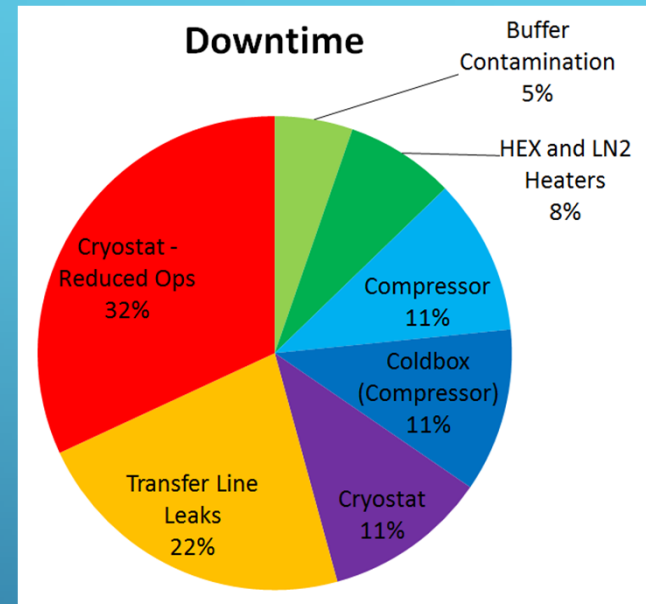
CESR-B HISTORY AT CLS

- ▶ Cavity 1 – delivered May 2003
 - ▶ Commissioned August 2003 in the storage ring
- ▶ Cavity 2 – delivered mid-2004
 - ▶ Cavity 1 removed from SR to commission Cavity 2
 - ▶ Small leak to insulation vac – reinstalled Cavity 1
- ▶ Cavity 1 leak to UHV – January 2013
 - ▶ Reinstalled Cavity 2
 - ▶ Cavity 1 to RI for repair – 27 months
 - ▶ Leaks at RF pickup flanges
 - ▶ Cavity 2 needed significant conditioning – 6 months



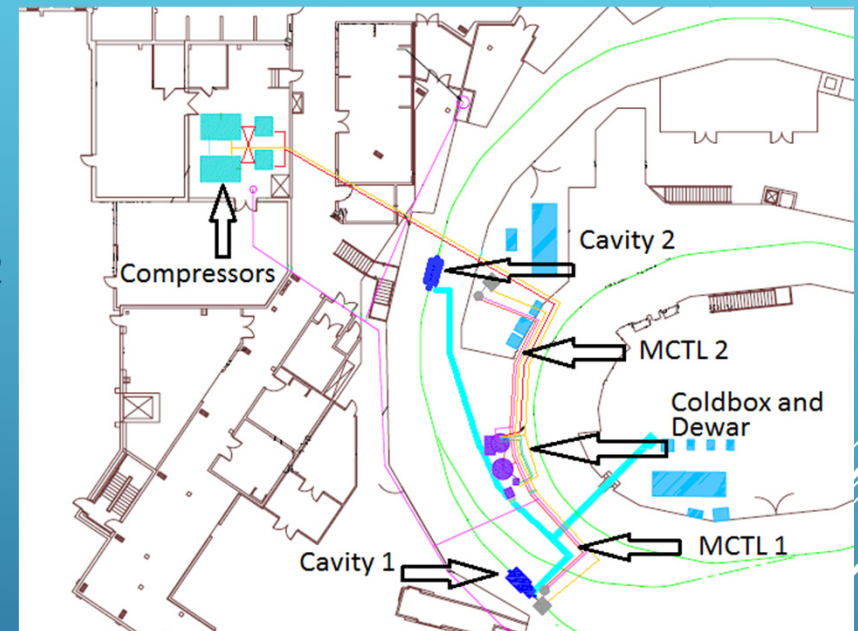
PAST PROBLEMS

- ▶ CLS is an “operating facility”
- ▶ Cryogenics at CLS is a “utility”
- ▶ SRF system has been generally reliable
- ▶ ~ 14 weeks of unplanned downtime
 - ▶ over a 13 year period
 - ▶ Continually striving for improvement



THE “CRYO UPGRADE”

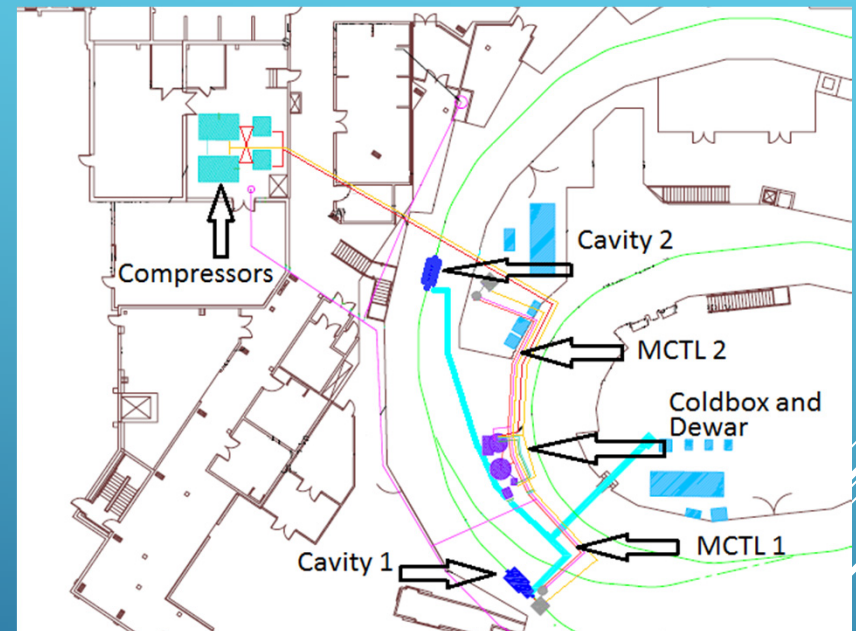
- ▶ Add spare cavity to ring
 - ▶ 2nd compressor inline
 - ▶ 2nd ORS/GMP
 - ▶ Move coldbox and dewar
 - ▶ New MCTLs and valve boxes
 - ▶ CLS spare module installed in straight 2
 - ▶ LN2 phase separators & control



THE “CRYO UPGRADE”

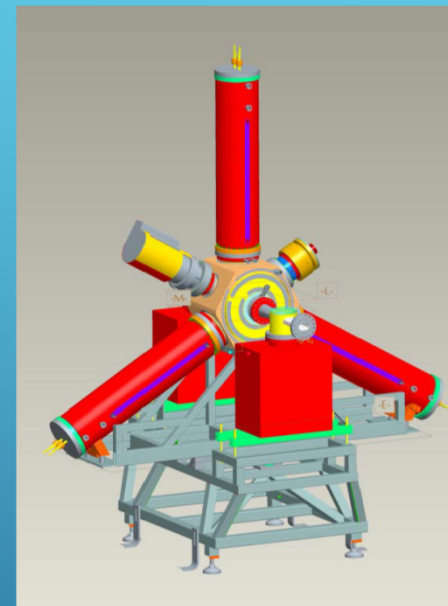
► Advantages

- Quick swap of compressors or ORS/GMP for maintenance or failure
- Elevated dewar and coldbox
- New shorter simpler MCTLs
- CLS spare module installed in straight 2
- LN2 phase separators & control



SUPERCONDUCTING VS. NORMAL-CONDUCTING

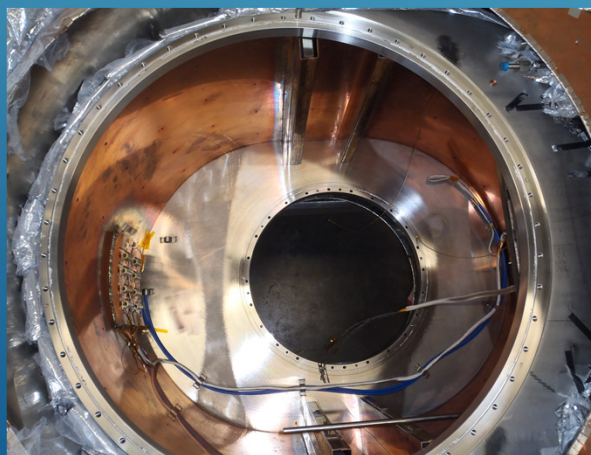
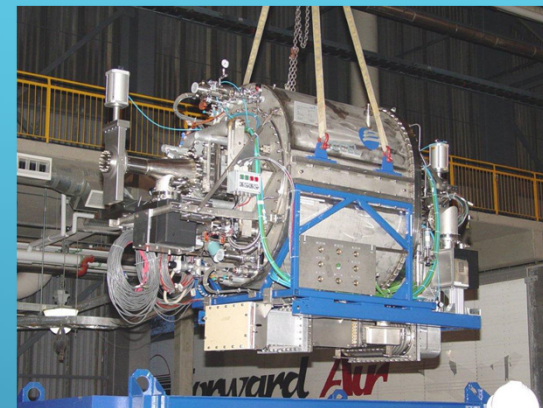
- ▶ Could we gain reliability by switching to a NC system
- ▶ Euro cavity (Alba, Bessy-II)
 - ▶ Better HOM damping than previous NC cavities
 - ▶ Still not as good as SC!
 - ▶ Easier to work on/repair
 - ▶ Lower voltage gradient – less arcing, multipactor etc.
- ▶ Decision: remain with SC
 - ▶ NC option very costly & time consuming
 - ▶ Cooling water upgrades, more RF transmitter power req'd
 - ▶ Longitudinal feedback required – complex system
 - ▶ The “devil you know” versus the “devil you don't”
- ▶ To reduce risk of failures SC would also require \$\$
- ▶ To become more self-reliant even more \$\$



Picture from: F. Perez et al. “New Developments for the RF System of the ALBA Storage Ring,” *Proc. EPAC 2006*, Edinburgh, Scotland.

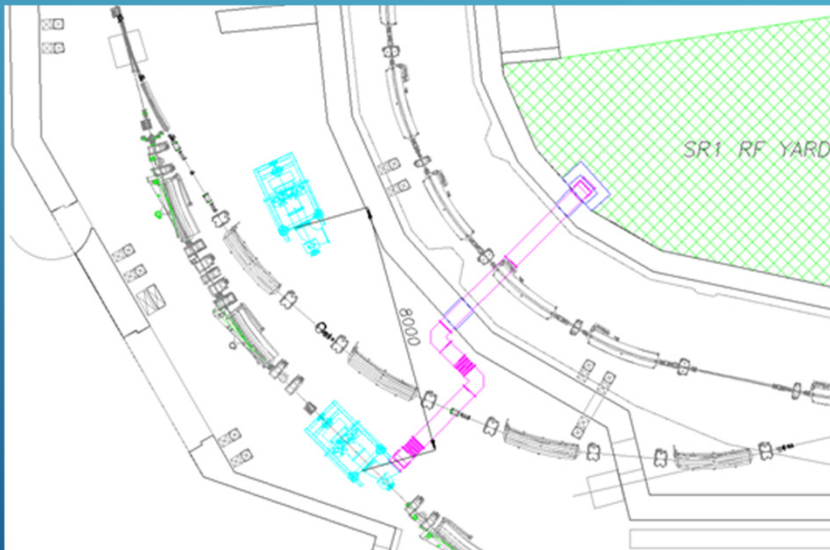
ANOTHER CESR-B FOR CLS

- ▶ Reduce risk due to double failure
 - ▶ Chance of double failure is statistically significant
 - ▶ Review of past failures indicates roughly 5-10% chance
 - ▶ Double failure upon installation with 27 month repair period = 27 months of downtime!
 - ▶ Risk of triple failure is considered acceptable
 - ▶ Roughly 0.25% to 1%



COLD RF TEST

- ▶ Connect spare cavity to cryo and instrumentation
 - ▶ Test RF and vacuum integrity when cold
 - ▶ Ensure it will function if required
 - ▶ Will occur during extended maintenance outage – Feb-Apr 2017



QUESTIONS?

